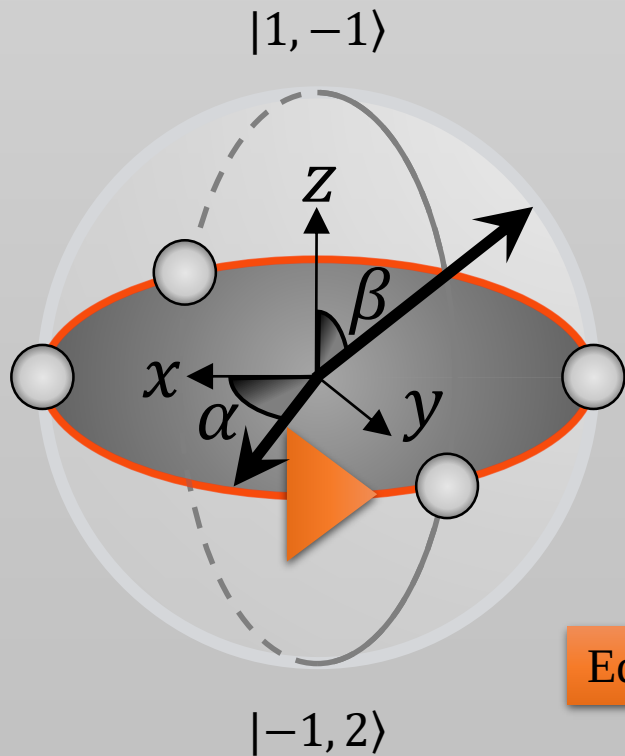
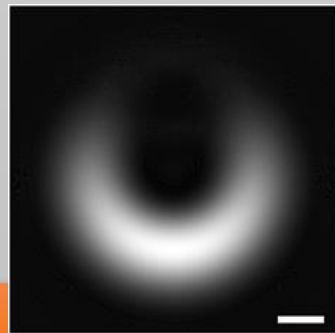


Gray exciton transition rate

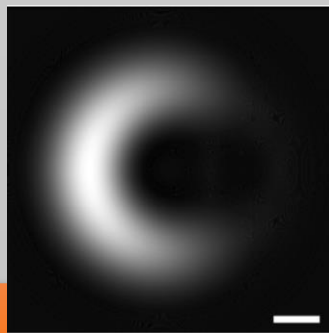
$$\Gamma_{G,\mathbf{Q}}^{1,-1,-1,2}(\mathbf{Q}; \alpha, \beta)$$



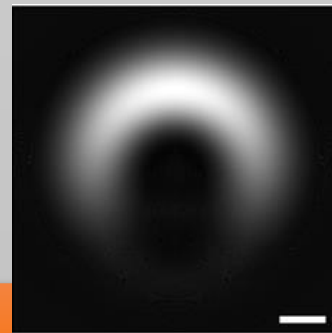
$\alpha = 0$



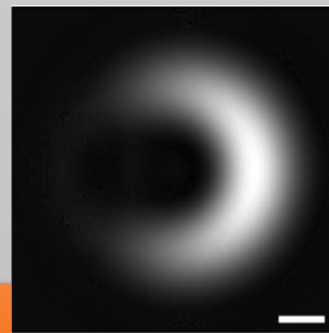
$\alpha = \pi/2$



$\alpha = \pi$



$\alpha = 3\pi/2$



Equator line $\beta = \pi/2$